

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560078

(An Autonomous Institution affiliated to VTU, Belagavi)

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Introduction to AI and Applications

CIA-1-Question Bank

Module 1(Complete) and Module 2(First half):

1. Define Artificial Intelligence (AI) and explain how AI works.
2. List and briefly explain the Advantages and Disadvantages of Artificial Intelligence.
3. Outline the History of Artificial Intelligence.
4. Demonstrate the differences between Weak AI (Narrow AI) and Strong AI (General AI) by applying them to real-world scenarios such as virtual assistants in customer service versus autonomous medical diagnosis systems.
5. Describe the four types of AI with suitable example for each: Reactive Machines, Limited Memory, Theory of Mind, and Self-Awareness.
6. Explain the differences, based on Transportation: Autonomous vehicles use AI to make real-time driving decisions between Artificial Intelligence, Augmented Intelligence, and Cognitive Computing.
7. Define Intelligence and list its key Components.
8. Apply the concept of human and machine intelligence to illustrate how each would approach evaluating student performance, and explain how their outcomes might differ in a classroom setting.
9. Demonstrate the concepts of an Agent and Environment by applying them to a real-world scenario such as a self-driving car navigating city traffic.
10. Differentiate between Machine Learning and Deep Learning and describe how they relate to the broader field of AI.
11. Discuss the purpose of AI and Ethical Concerns that arise with the increasing adoption of AI technologies.
12. Explain the significance of Moral Issues and Data Privacy as AI trends

13. What is AI Bias? Provide an example of how it can manifest in an application

14. Identify and briefly discuss two or three Recent trends in AI